Homework 6

Exercise List

Undergrad CMOS Course – 2026
Fundación FULGOR
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Design Guidelines

For this assignment, the circuit designs will use the characterization curves covered in class and the real PDK sky130A. The exercises are similar to those in Homework 5, adapted for use with the design tool. Therefore, the general assumptions used in Homework 5 no longer apply; use the device models provided by the PDK. Use the PDK devices `nfet_01v8` and `pfet_01v8` for all transistors throughout the assignment.

The maximum allowed width per finger W is $10\,\mu$m. Widths specified in the exercises are required equivalent widths, W_{eq}. If a larger equivalent width is required, use the transistor parameters *number of fingers* (n_f) and/or *multipliers* (m), where

$$W_{\text{eq}} = W\, m\, n_f.$$

Different combinations of W, m, and n_f can produce different results even when they have the same W_{eq} (why?).

Part I — Exercises

Exercise 1 — Problem 7.2

Design the circuit shown in Fig. 7.40 for a drain current of 1 mA. If $W/L = 20/0.18$, calculate R_1 and R_2 such that the input impedance is at least $20\,\text{k}\Omega$. Measure the input impedance using an AC analysis at the input node (the gate of M_1) at low frequencies.

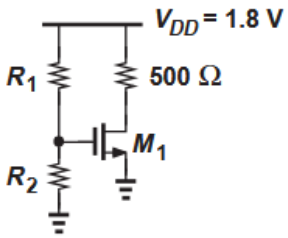


Figure 7.40

Exercise 2 — Problem 7.4

The circuit shown in Fig. 7.42 must be designed for a voltage drop of 200 mV across R_S.

(a) Calculate the minimum allowable value of W/L if M_1 must remain in saturation.

(b) Calculate the required values of R_1 and R_2 if the input impedance must be at least $30\,\text{k}\Omega$. Measure the input impedance using an AC analysis at the input node (the gate of M_1) at low frequencies.

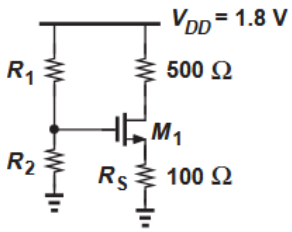


Figure 7.42

Exercise 3 — Problem 7.6

The self-biased stage shown in Fig. 7.44 must be designed for a drain current of 1 mA. If M_1 must provide a transconductance of $1/(100\,\Omega)$, calculate the required value of R_D. Neglect R_G.

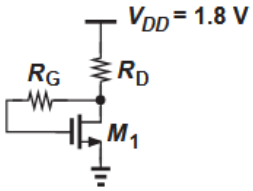


Figure 7.44

Exercise 4 — Problem 7.17

In the common-source stage shown in Fig. 7.54, $W/L = 30/0.18$.

(a) What gate voltage yields a drain current of 0.5 mA? Verify that M_1 operates in saturation.

(b) With such a drain bias current, calculate the low-frequency voltage gain of the stage.

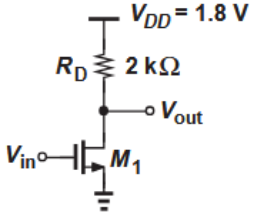


Figure 7.54

Exercise 5 — Problem 7.20

The common-source stage shown in Fig. 7.56 must provide a low-frequency voltage gain of 10 with a bias current of 0.5 mA. The tool does not provide the parameter r_o; calculate it as $r_o = 1/g_{ds}$. Thus,

$$A_v = g_{m1}(r_{o1} \parallel r_{o2}) = \frac{g_{m1}}{g_{ds1} + g_{ds2}}.$$

(a) Calculate the required value of $(W/L)_1$.

(b) If $(W/L)_2 = 20/0.18$, calculate the required value of V_B.

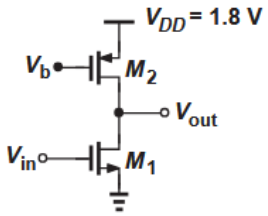


Figure 7.56

Exercise 6 — Problem 7.25

Design the circuit shown in Fig. 7.59 for a voltage gain of 3. If $(W/L)_1 = 20/0.18$, determine $(W/L)_2$.

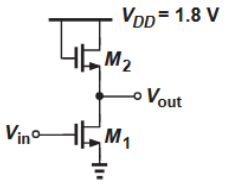


Figure 7.59